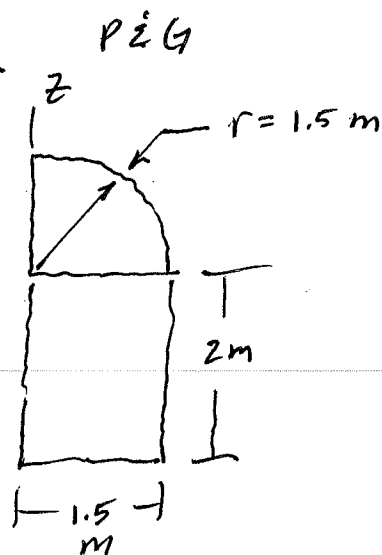


(Hibbeler)
F9-16



Find: VOLUME
SURFACE AREA

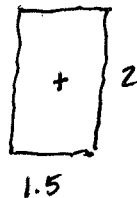
Full Rotation = 360°
about z axis

VOLUME



$$A = \frac{1}{4} \pi (1.5)^2 = 1.767 \text{ m}^2$$

$$\bar{r} = \frac{4R}{3\pi} = \frac{4(1.5)}{3\pi} = \frac{6}{3\pi} = \frac{2}{\pi} = .6366 \text{ m}$$



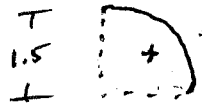
$$A = 1.5(2) = 3 \text{ m}^2$$

$$\bar{r} = \frac{1.5}{2} = .75 \text{ m}$$

$$VOL = \sum \theta \bar{r} A = 2\pi [.6366(1.767) + .75(3)]$$

$$VOL = 21.2 \text{ m}^3$$

SURFACE AREA



$$L = \frac{2\pi(1.5)}{4} = \frac{3\pi}{4}$$

$$\bar{r} = \frac{2R}{\pi} = \frac{2(1.5)}{\pi} = \frac{3}{\pi}$$

$$\sim L = 2, \bar{r} = 1.5$$

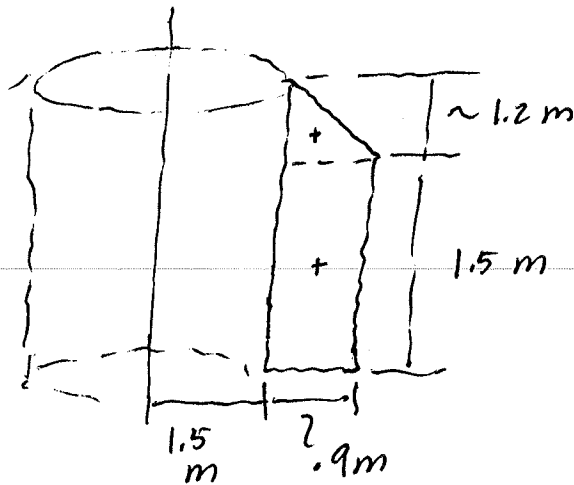
$$L = 1.5, \bar{r} = .75$$

$$A = 2\pi \left[\frac{3\pi}{4} \cdot \frac{3}{\pi} + 1.5(2) + .75(1.5) \right]$$

$$A = 40.06 \text{ m}^2$$

P&G Example (Hibbeler F9-14)

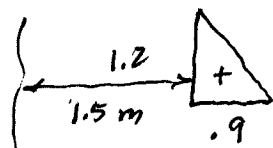
F9-14 z



Find Surface Area
and Volume ...

360° Rotation ..
about z axis

Volume



$$A = \frac{1}{2}(.9)(1.2) = .540 \text{ m}^2$$

$$\bar{r} = 1.5 + .3 = 1.8 \text{ m}$$



$$A = .9(1.5) = 1.35 \text{ m}^2$$

$$\bar{r} = 1.5 + .45 = 1.95 \text{ m}$$

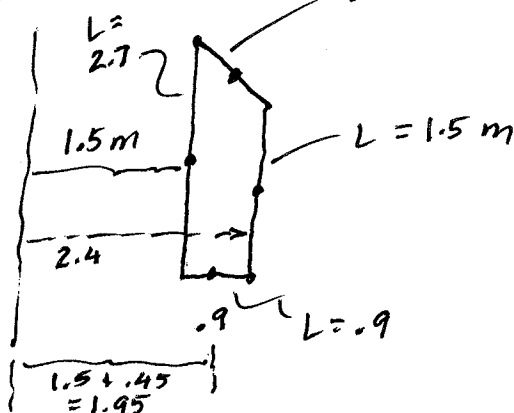
$$\theta = 2\pi$$

$$VOL = \sum \theta \bar{r} A = 2\pi [1.8(.54) + 1.95(1.35)]$$

$$VOL = 22.65 \text{ m}^3$$

Surface Area

$$L = \sqrt{.9^2 + 1.2^2} = 1.5 \text{ m}$$



$$A = \sum \theta \bar{r} L$$

$$= 2\pi [1.5(2.7) + (1.95)(1.5) + 2.4(1.5) + (1.95)(.9)]$$

$$A = 77.47 \text{ m}^2$$